

M002 Week 5 — English Worksheet

Topic: Redstone Basics — Signals and Power · **Course:** Maze Madness · **Time:** about 45 minutes

This week your agent meets a **closed redstone door**. The door blocks the path — so the agent treats it like a wall and goes around it.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
keep doing forever:
  if wall ahead:
    turn right
  otherwise:
    move forward
```

A closed door is in front of the agent. Does the agent treat it as a wall? Why?


```
keep doing forever:
  if wall ahead OR door ahead:
    turn right
  otherwise:
    move forward
```

How is this different from the code above? What does it do at a closed door?

A maze has:

- one open path
- one closed door blocking a shortcut

Should the agent ever try to walk through the door? Why or why not?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to treat **both walls and closed doors** as obstacles.

```
if wall ahead:
    turn right
otherwise:
    move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to **go around** the closed door by turning, walking past, then turning back.

```
if door ahead:
    move forward
```

Hint: moving forward into a door doesn't work. Write a route-around instead.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A redstone block next to a door **opens** it. The agent should stop treating that door as a wall once redstone is placed.

```
if door ahead:
    turn right
```

Hint: add a check — only turn if the door is **closed**.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. The maze has at least one **closed redstone door** in the way. Write code that **routes around** the door to reach the goal. When you finish, come back here.

Send a photo or video of the agent reaching the end, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I read the maze and noticed ...

I treated the door as a wall because ...

The agent went around the door by ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while the agent runs the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **redstone**, **door**, **signal**, **wall**, **route**.

1. Show the closed door in the maze.
2. Run your code and show the agent going around it.
3. Read your code out loud and say which line stops the agent walking into the door.
4. Show one bug you hit and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.